

Question Number	Answer	Additional guidance	Mark
2(a)(i)	The only correct answer is D animal, plant and prokaryotic cells <i>A is not correct because all three cell types contain a cell membrane</i> <i>B is not correct because all three cell types contain a cell membrane</i> <i>C is not correct because all three cell types contain a cell membrane</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(a)(ii)	The only correct answer is C plant and prokaryotic cells only <i>A is not correct because prokaryotic cells contain a cell wall</i> <i>B is not correct because animal cells do not contain a cell wall</i> <i>D is not correct because animal cells do not contain a cell wall</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(b)(i)	A calculation in which: - cell diameter measured and converted into micrometres (1) - diameter divided by 25 with no units (1) OR 25 converted to mm - diameter divided by 0.025 with no units	<u>Example of calculation:</u> 42mm=42000 µm 42000 ÷ 25 = x1680 ecf for ±1mm and incorrect conversions Correct answer with no units and with no working scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that includes one of the following points: (site of) photosynthesis / converts light energy to {chemical energy / ATP} (1)	ACCEPT formation of correct named product of photosynthesis e.g. glucose ignore makes food	(1)

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	An answer that includes the following point: this organism has a nucleus / prokaryotic cells do not contain a nucleus (1)	DO NOT ACCEPT cell membrane / cytoplasm ignore contains membrane bound organelles	(1)

Question Number	Answer	Additional guidance	Mark
2(c)(i)	• plasmodesma / plasmodesmata (1)		(1)

Question Number	Answer	Additional guidance	Mark
2(c)(ii)	An explanation that includes the following points: • communication between / connects (connected) cells (1) • therefore signalling substances pass through {symplast / cytoplasm} / cytoplasmic streaming / transport of suitable named molecules e.g. minerals, water, glucose, amino acids, proteins, RNA (1)	allow ECF for pits, cell wall and middle lamella only pits • communication between / connects (connected) cells • therefore transport of suitable named molecules cell wall • (sieve like network made of) cellulose fibres support the cell • therefore helps retain rigid structure / prevent cell lysis middle lamella • joins (adjacent) cell (walls) together • therefore increases {strength / stability} of {plant / cell (wall)}	(2)